A Program in a Container

Packaging software for predictable runs

SCAN / LEARN

THE MAIN IDEA

A container packages an application with its runtime dependencies. Containers share the host operating-system kernel; they are not automatically separate virtual machines.

FOLLOW THE SYSTEM

01

APPLICATION

A program needs specific libraries.

›

02

IMAGE

Code and dependencies are packaged.

›

03

CONTAINER

The image runs as an isolated process.

›

04

HOST

The host supplies shared kernel and resources.

CODE + LIBRARIES → IMAGE → CONTAINER

COMMON MISCONCEPTION

A container includes an entire independent operating system.

MORE ACCURATE

Isolation is limited; the host kernel is still shared.

THREE QUESTIONS TO KEEP

PROBLEM: How can software run predictably elsewhere?

FAILURE: Missing dependencies or bad permissions break it.

IN REAL LIFE: Services, development and deployments.

SOLO DISCOVERY

List what an app needs beyond its code to start.